

Wazuh SIEM with File Integrity Monitoring

Documentation

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Table of Contents

1. Overview
2. Lab Architecture
3. Prerequisites
4. Installing the Wazuh Manager (Ubuntu)
 - a. 4.1 Add Wazuh GPG Key
 - b. 4.2 Download and Execute Wazuh Installation Script
5. Accessing the Wazuh Dashboard
6. Registering the Agent with the Manager
 - a. 6.1 Generate Agent Key on Ubuntu Manager
 - b. 6.2 Apply Key in the Windows Agent
7. File Integrity Monitoring on Windows
 - a. 7.1 Edit Agent Configuration
 - b. 7.2 Restart the Agent
8. Verifying Setup

1. Overview

This project implements a **Security Information and Event Management (SIEM)** environment using **Wazuh**.

The Wazuh Manager is installed on **Ubuntu 22.04.5 LTS**, while a **Windows endpoint** runs the Wazuh Agent. The agent collects security information from the Windows system and sends it to the Wazuh Manager for analysis.

File Integrity Monitoring (FIM) is also configured to detect changes such as file creation, modification and deletion on the Windows endpoint.

2. Lab Architecture

Components

Component	Purpose
Ubuntu 22.04.5 LTS	Wazuh Manager
Windows	Monitored endpoint
Wazuh Agent 4.14.7	Collects endpoint information
Wazuh Dashboard	Security event visualization
FIM	Monitors file changes

Architecture Flow

Your architecture image can be placed here:

architecture/
└── wazuh-siem-fim-architecture.png

3. Prerequisites

Ubuntu

- Ubuntu 22.04.5 LTS
- Internet connection
- Sudo/root privileges
- Minimum hardware requirements for Wazuh

Check Ubuntu version:

`lsb_release -a`

Expected:

Distributor ID: Ubuntu

Description: Ubuntu 22.04.5 LTS

Release: 22.04

Codename: jammy

Windows

- Windows endpoint
- Administrator PowerShell
- Wazuh Agent MSI installer
- Network connectivity to Ubuntu Manager

In your implementation:

Wazuh Manager IP: 192.168.1.11

Wazuh Agent: 4.14.7

4. Installing the Wazuh Manager (Ubuntu)

4.1 Add Wazuh GPG Key

The GPG key allows Ubuntu to verify packages obtained from the Wazuh repository.

Run:

```
sudo apt update
```

Install required packages:

```
sudo apt install curl gnupg apt-transport-https unzip -y
```

Add the Wazuh GPG key:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | \  
sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

If the key already exists and Ubuntu asks:

```
File exists. Overwrite? (y/N)  
enter:y
```

4.2 Download and Execute Wazuh Installation Script

Download the Wazuh 4.14 installation assistant:

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
```

Check that the file exists:

```
ls -lh wazuh-install.sh
```

Give execution permission:

```
chmod +x wazuh-install.sh
```

Run the all-in-one installation:

```
sudo bash ./wazuh-install.sh -a
```

The **-a** option performs the all-in-one installation of the Wazuh central components. The official quickstart currently documents the same 4.14 installation-assistant approach. ([Wazuh Documentation](#))

During installation, the system generates certificates and installs the Wazuh components.

After successful installation, the output provides the dashboard URL and administrator credentials.

To retrieve the generated passwords:

```
sudo tar -O -xvf wazuh-install-files.tar \
wazuh-install-files/wazuh-passwords.txt
```

5. Accessing the Wazuh Dashboard

Find the Ubuntu IP address:

```
ip addr
```

For your setup, the Manager IP was:

```
192.168.x.x
```

Open a browser on the Windows system: `https://manager_ip`

Log in using:

Username: admin

Password: <generated password>

The Wazuh dashboard provides visualization of security events, alerts and agent information. The first connection may show a certificate warning because the installation uses generated certificates. ([Wazuh Documentation](#))

Screenshot to add:

 Wazuh Dashboard login/dashboard

6. Registering the Agent with the Manager

Agent enrollment registers the Windows endpoint with the Wazuh Manager and establishes authenticated communication between them. ([Wazuh Documentation](#))

6.1 Generate Agent Key on Ubuntu Manager

Open Ubuntu Terminal.

Run:

```
sudo /var/ossec/bin/manage_agents
```

Select: A to add an agent.

Enter:

Name: Windows-Agent

IP Address: any

The manager generates an agent ID.

Then select: E to extract the key.

Select the ID of the Windows agent.

Copy the generated key.

You can also list registered agents using:

```
sudo /var/ossec/bin/manage_agents -l
```

The `manage_agents` utility is specifically provided for managing Wazuh agent authentication keys. ([Wazuh Documentation](#))

6.2 Apply Key in the Windows Agent

Open **PowerShell as Administrator**.

Your Wazuh Agent was installed in:

```
C:\Program Files (x86)\ossec-agent
```

The Wazuh configuration file is:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

This is the documented Windows agent location for a 64-bit installation. ([Wazuh Documentation](#))

Check the agent service:

```
Get-Service WazuhSvc
```

Start the agent:

```
Start-Service WazuhSvc
```

Verify:

Get-Service WazuhSvc

Expected:

Status	Name	DisplayName
Running	WazuhSvc	Wazuh

7. File Integrity Monitoring on Windows

File Integrity Monitoring (FIM) monitors files and directories for changes. Wazuh can detect events such as **file creation, modification and deletion** and report them to the Wazuh Manager. ([Wazuh Documentation](#))

7.1 Edit Agent Configuration

Create a test directory on Windows:

```
New-Item -ItemType Directory -Path "C:\Users\SKY\Desktop\Wauzh"
```

Open the Wazuh configuration file:

```
notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
```

Inside the <syscheck> section, add:

```
<syscheck>  
  <directories realtime="yes">C:\SKY\Desktop\Wauzh </directories>  
</syscheck>
```

7.2 Restart the Agent

After saving `ossec.conf`, restart the Wazuh service:

Restart-Service WazuhSvc

Check the service:

Get-Service WazuhSvc

Expected:

Running WazuhSvc Wazuh

8. Verifying Setup

8.1 Test File Creation

Create a test file:

New-Item "C:\SKY\Desktop\Wauzh \test.txt"

Expected:

test.txt

8.2 Test File Modification

Add content:

**"FIM test - file modified" | Out-File "C:\SKY\Desktop\Wauzh \test.txt" -
Append**

8.3 Test File Deletion

Remove-Item "C:\ SKY\Desktop\Wauzh \test.txt"

8.4 Verify Wazuh Dashboard

Open: [https://](https://manager_ip) manager_ip

Go to the Wazuh dashboard and check the security events/alerts generated by the Windows agent.

The FIM events should show changes associated with the monitored directory.